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B. E. (Eighth Semester)

EXAMINATION, April-May, 2015

(Old Scheme)

(Branch : Et & T)

RADAR ENGINEERING AND NAVIGATIONAL AIDS

(Elective-III)

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks 28*

Note : (i) Part (a) carry 2 marks.

(ii) Attempt any *two* parts from part (b), (c) and (d) from each question.

Unit—I

1. (a) Define Radar cross-section of target. 2

(b) Derive Radar range equation and mention its significance. 7

(c) Explain False alarm and derive the expression for probability of false alarm. 7

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(d) Explain the block diagram of Pulse Doppler Radar. 7

Unit—II

2. (a) Explain Doppler effect with proper mathematical expression. 2

(b) Explain the block diagram of MTI radar with power oscillator transmitter. 7

(c) Explain the lobe switching and conical scanning technique used in radar. 7

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(d) Explain the working of Continuous wave radar using Doppler information. 7

Unit—III

3. (a) List major effects of propagation or radar performance. 2

(b) What radar characteristics are important for detecting target at low altitudes ? 7

(c) Explain the effect of attenuation by atmospheric gases. 7

(d) Explain what effect does the local oscillator have on the receiver dynamic range ? 7

Unit—IV

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4. (a) Why does a parabolic surface make a good reflector antenna ? 2

(b) Describe two methods for feeding a paraboloid reflector. Under what condition is the methods of feed unsatisfactory ? 7

(c) What are the advantage and limitations of operating a ground-base air surveillance radar under a radome. Compare the air-inflated radome and the rigid geodesic redome for this application. 7

(d) Define directive gain and power gain of antenna. 7

Unit—V

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5. (a) What is Duplexer ? 2

(b) Why might a double conversion superhetrodyne receiver be used instead of single conversion receiver ? Explain using proper block diagram. 7

(c) Explain different types of mixer used in superheterodyne receiver. 7

(d) Compare the advantages and disadvantages of a solid-state transmitter, klystron transmitter, and magnetron transmitter. 7

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